

# UJWAL U

Full-Stack Developer | AI/GenAI & RAG Engineer

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## PROFESSIONAL SUMMARY

3rd-year B.Tech CSE student (BMSIT, Bangalore, GPA 8.24) and full-stack developer with hands-on experience across the stack - REST APIs, relational and vector databases, and AI-native/RAG pipelines. Currently building production backend services as a Web Developer Intern at Rail Wheel Factory, and recently architected a 700K-chunk hybrid retrieval backend at Hacker House Goa 2026. Comfortable across Java, Python, JavaScript/TypeScript, and SQL, with production experience in Node.js/Express, FastAPI, React, and Next.js. As Chair of the IEEE EMBS Student Branch Chapter, I also lead technical community-building, event operations, and grant-funded initiatives at scale. Strong fundamentals in DSA, OOP, and distributed systems design, with a genuine passion for building AI-powered products end to end.

## EXPERIENCE

### Web Developer Intern - Rail Wheel Factory (RWF), IT Department

Jul 2026 - Present | Bangalore

- Built the backend for Campus Infrastructure Reservation System (CIRS) - REST APIs and relational schema in Node.js, Express.js, and MySQL for resource booking and allocation.
- Engineered backend logic for PrepMind, an AI mock-interview platform - Supabase/Postgres schema design, Clerk auth, and Groq LLM API integration for automated interview scoring.
- Optimized database queries and API response times across production RWF applications.

### GenAI Intern - Augsta Infosystem Pvt. Ltd.

Jun - Nov 2025 | Bangalore

- Designed and deployed enterprise RAG pipelines and LLM-backed agents with real-time data retrieval.
- Built backend workflow-automation agents using Supertool orchestration and the Serper API for live web search.
- Applied production-grade prompt engineering and pipeline optimization across multiple GenAI deployments.

## PROJECTS

### Personal Portfolio Website

Personal Project

- Full-stack portfolio site (React, Node.js/Express, MongoDB Atlas) with a resume-download feature and a Nodemailer-powered contact form.
- Deployed on Vercel with a Cloudinary-ready image pipeline for project media.

### FaceTrace - Face Identification & Blockchain Verification

Personal Project

- Built an end-to-end pipeline that detects and encodes a face from a scanned image, then runs a dynamic (non-hardcoded) web/social-media search to surface a genuine matching post.
- On a verified match, uploads the post's cryptographic hash/fingerprint to a blockchain, creating a tamper-evident, publicly verifiable identity record.

### Voice-Enabled Multilingual RAG System

Hacker House Goa 2026 - Task 2

- Architected a hybrid-retrieval backend supporting 14 Indic languages + English: FAISS index over 700K chunks, with BM25 + dense retrieval fused via Reciprocal Rank Fusion, served through FastAPI.
- Integrated Sarvam AI STT/TTS for voice I/O and Groq (Llama 3.1 8B Instant) for low-latency generation; deliberately avoided LangChain and hosted vector DBs to keep the retrieval stack lean and self-managed.
- Next.js frontend, Sentence Transformers for embeddings.

### Builder ID Card Generator

Hacker House Goa 2026 - Task 1

- Backend-driven photo-frame/ID generator using Upstash Redis for atomic unique-ID issuance and Supabase for persistence, with Next.js API routes.
- Built end-to-end using Google Antigravity, a Gemini-powered agentic IDE.

### PrepMind - AI Mock Interview Platform

Personal Project

- Full-stack placement-prep platform: Next.js frontend, Supabase/Postgres backend, Clerk auth, Groq API for AI-driven interview simulation, Chart.js analytics.
- Diagnosed and resolved production backend issues — RLS policy failures via service-role key, credential rotation, and deployment fixes on Vercel.

### Smart Foot - Piezoelectric Energy Harvesting

Sustainathon 2025 - 3rd Place + ₹10,000

- Prototype converting footsteps into electrical energy; React + Supabase live dashboard for real-time monitoring.
- Tech: React, Supabase, ESP32, Piezoelectric Sensors

### Blue Carbon Ledger - Blockchain Carbon Registry

Smart India Hackathon 2025 - Round 2

- Decentralised MRV platform for mangrove/wetland carbon credits on Polygon with IPFS off-chain storage.
- Tech: React, TypeScript, Solidity, Polygon, IPFS, Ethers.js

### Soil Monitoring System

HackSphere - 3rd Place + Internship Offer

- Real-time soil health monitoring using IoT sensors, awarded under Sustainability & Social Impact.
- Tech: IoT Sensors, Hardware Prototyping

### Robotics Workshop

IEEE RAS Summer of Projects - Top 5 of 40 Teams

- Built line-following, gesture-controlled robots; MIT App Inventor mobile app for wireless robot control.
- Tech: ESP32, Arduino IDE, Infrared sensors

## SKILLS

**Languages:** Java, C++, Python, JavaScript (ES6+), TypeScript, SQL

**Backend:** Node.js, Express.js, FastAPI, REST APIs, Supabase/PostgreSQL, MySQL, Redis (Upstash)

**AI/GenAI:** RAG Pipelines, FAISS, Hybrid Retrieval (BM25 + Dense), LLM APIs (Groq), Prompt Engineering

**Frontend:** React.js, Next.js, Tailwind CSS

**Tools:** Git, GitHub, Postman, VS Code

## EDUCATION

### B.E. Computer Science & Engineering - BMS Institute of Technology & Management

2024 - 2028

GPA: 8.24 / 10 | Bangalore

12th Grade - CBSE - 78%

2024

10th Grade - ICSE - 86%

2022

## LEADERSHIP & EXTRACURRICULARS

- Chair - IEEE Engineering in Medicine & Biology Society (EMBS), BMSIT
- Vice Head, Media & Marketing - Entrepreneurship Cell (E-Cell), BMSIT
- Vice President - NSS Unit, BMSIT
- Vice Head, Marketing - Coding Club, BMSIT
- Web Dev Team - Alterino, BMSIT
- Events & Operations - IEEE Signal Processing Society (SPS), BMSIT
- Operations Team - Sports Club, BMSIT

(Jan 2025 - Present)

(May 2025 - Present)

(May 2025 - Present)

(Apr 2025 - Jan 2026)

(Mar 2025 - Jan 2026)

(Jan 2025 - Jan 2026)

(May 2025 - Nov 2025)